Project Name: Flexible Object Detection System
(Dynamic Addition of New Objects Without Retraining)

Team: CODENOVA

Problem Statement:-

Design an object detection system that can recognize new objects without requiring complete retraining of the model, using techniques like fine-tuning or transfer learning.

Members:

- Priyesh Shrivastav(Leader)
- Louv Krishna Upadhyay
- Rajnish Kumar Singh
- Sharad Pal
- Rishikesh

Github link: *https://github.com/louvkrishnaupadhyay/Code-Ronin-Hackathon*

Problem Statement:-

- **Traditional object detection systems require complete retraining to add new objects**

- **Retraining is:**
  ~Time-consuming
  ~ Computationally expensive

- **Systems cannot adapt to:**
  ~New or unseen objects
  ~Dynamic real-world environments

- **Not suitable for:**
  ~Real-time applications
  ~Accessibility solutions

Solution Overview:-

- Developed a system that can add and recognize new objects without full retraining

- Uses:
  - ~ YOLO for object detection
  - ~ DINOv2 for feature extraction
  - ~ FAISS for similarity-based matching
  - ~ Quantum-enhanced similarity using quantum circuits (PennyLane)

- Converts objects into vector embeddings

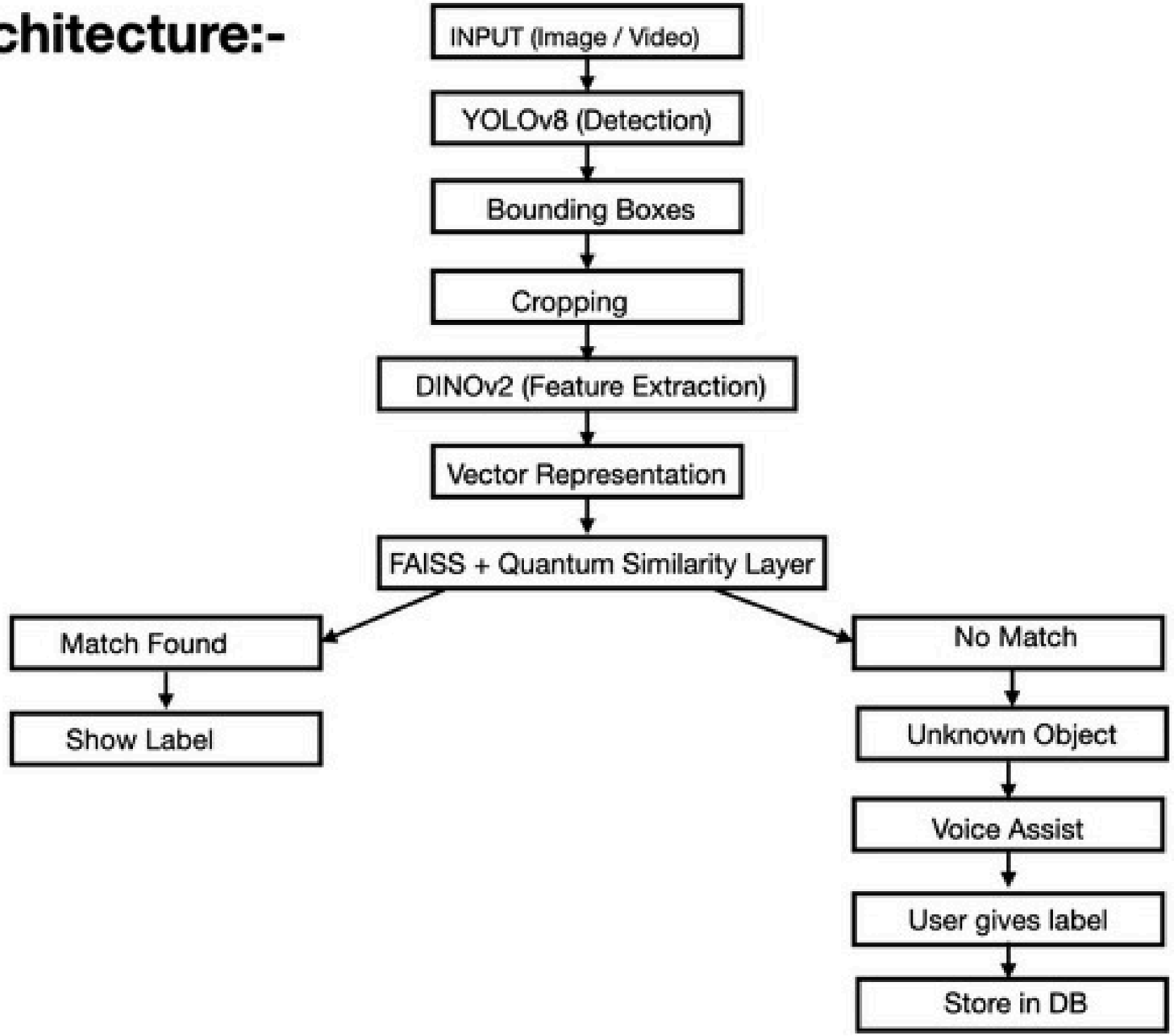
- New objects are learned instantly by storing their embeddings

- Works in real-time and adapts dynamically

Quantum Enhancement:

- Introduces Quantum Machine Learning into the system
- Feature vectors are encoded into quantum states (qubits)
- Quantum circuits are used to compute similarity between objects
- Enhances pattern recognition using superposition and entanglement
- Improves detection of new/unseen objects without retraining
- Implemented using PennyLane (quantum simulation framework)

System Architecture:-



Tech Stack:

Frontend:

HTML5 + CSS + JavaScript

Streamlit

Backend/AI:

Python

OpenCV

YOLOv8

DINOv2

NumPy

PennyLane (Quantum ML)

Storage:

Local JSON / Pickle

OR

SQLite

Real-World Impact:-

- **Security & Surveillance**
  - ~ Detect unknown or suspicious objects in real-time
  - ~ Enhance public safety and threat detection

- **Accessibility**
  - ~ Voice-assisted object recognition
  - ~ Helps visually impaired users navigate surroundings

- **Industry & Manufacturing**
  - ~ Track new objects without retraining delays
  - ~ Improves efficiency and reduces downtime

- **Smart Systems**
  - ~ Adaptive detection in autonomous systems
  - ~ Handles dynamic environments effectively

Innovation + Closing:-

Innovation:

- No need for complete model retraining
- New objects can be added instantly
- Uses vector-based matching instead of training
- Separates detection and recognition modules
- Integrates Quantum Machine Learning for enhanced similarity matching

Key Features:

- Real-time performance
- No catastrophic forgetting
- Scalable and flexible system
- Works efficiently in dynamic environments

Conclusion:

- Provides a smarter and adaptive object detection system
- Reduces time, cost, and computational effort
- Suitable for real-world applications